STAGE 1 — Install tools

```
keerthi@KeerthiVarma:~$ sudo apt update
sudo apt install -y tshark iperf3 mtr-tiny iproute2 wireshark netcat-openbsd
[sudo: authenticate] Password:
Hit:1 http://archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
76 packages can be upgraded. Run 'apt list --upgradable' to see them.
netcat-openbsd is already the newest version (1.234-1).
netcat-openbsd set to manually installed.
Upgrading:
  iproute2  libpcre2-8-0

Installing:
  iperf3  mtr-tiny  tshark  wireshark

Installing dependencies:
  alsa-topology-conf       libhwloc-plugins    libpcap0.8t64            libsm6              libwayland-server0
  alsa-ucm-conf            libhwloc15          libpcre2-16-0            libsmi2t64          libwebp7
  gnome-themes-extra-data  libibverbs1         libpgm-5.3-0t64          libsnappy1v5        libwebpmux3
  ibverbs-providers        libice6             libpipewire-0.3-0t64     libsndfile1         libwebrtc-audio-processing-1-3
  libaacs0                 libicu78            libpipewire-0.3-common   libsoxr0            libwireshark-data
  libabsl20260107          libinput-bin        libprotobuf32t64         libspa-0.2-modules  libwireshark19
  libasound2-data          libinput10          libprotoc32t64           libspandsp2t64      libwiretap16
  libasound2t64            libiperf0           libproxy1v5              libspeex1           libwsutil17
  libasyncns0              libjbig0            libpulse0                libspeexdsp1        libx264-165
  libavcodec62             libjpeg-turbo8      libqt6core5compat6       libsrt1.5-gnutls    libx265-215
  libavformat62            libjpeg8            libqt6core6t64           libssh-4            libxcb-cursor0
  libavutil60              libjsoncpp26        libqt6dbus6              libsvtav1enc2       libxcb-icccm4
  libb2-1                  liblerc4            libqt6gui6               libswresample6      libxcb-image0
  libbcg729-0              liblua5.4-0         libqt6multimedia6        libswscale9         libxcb-keysyms1
  libbdplus0               libmaxminddb0       libqt6network6           libtbb12            libxcb-render-util0
  libbluray3               libmbedcrypto16     libqt6opengl6            libtbbbind-2-5      libxcb-shape0
```

```
keerthi@KeerthiVarma: ~
Setting up libqt6network6:amd64 (6.10.2+dfsg-7) ...
Setting up libpcap0.8t64:amd64 (1.10.6-1ubuntu1.1) ...
Setting up libpipewire-0.3-0t64:amd64 (1.6.2-1ubuntu1.2) ...
Setting up qt6-qpa-plugins:amd64 (6.10.2+dfsg-7) ...
Setting up libavformat62:amd64 (7:8.0.1-3ubuntu2) ...
Setting up libqt6qml6:amd64 (6.10.2+dfsg-3) ...
Setting up libqt6opengl6:amd64 (6.10.2+dfsg-7) ...
Setting up libqt6qmlmodels6:amd64 (6.10.2+dfsg-3) ...
Setting up libqt6widgets6:amd64 (6.10.2+dfsg-7) ...
Setting up libqt6qmlworkerscript6:amd64 (6.10.2+dfsg-3) ...
Setting up libqt6svg6:amd64 (6.10.2-2) ...
Setting up libwireshark19:amd64 (4.6.4-1) ...
Setting up libqt6waylandclient6:amd64 (6.10.2+dfsg-7) ...
Setting up qt6-svg-plugins:amd64 (6.10.2-2) ...
Setting up libqt6qmlmeta6:amd64 (6.10.2+dfsg-3) ...
Setting up libqt6wlshellintegration6:amd64 (6.10.2+dfsg-7) ...
Setting up wireshark-common (4.6.4-1) ...
Setting up libqt6printsupport6:amd64 (6.10.2+dfsg-7) ...
Setting up tshark (4.6.4-1) ...
Setting up libqt6quick6:amd64 (6.10.2+dfsg-3) ...
Setting up libqt6multimedia6:amd64 (6.10.2-2) ...
Setting up wireshark (4.6.4-1) ...
Setting up libqt6waylandcompositor6:amd64 (6.10.2-4) ...
Setting up qt6-wayland:amd64 (6.10.2-4) ...
Processing triggers for libc-bin (2.43-2ubuntu2.4) ...
Processing triggers for man-db (2.13.1-1build1) ...
Processing triggers for libselinux1:amd64 (3.9-4build1) ...
Processing triggers for shared-mime-info (2.4-5build3) ...
Processing triggers for udev (259.5-0ubuntu3.4) ...
Processing triggers for hicolor-icon-theme (0.18-2build1) ...
keerthi@KeerthiVarma:~$
```

STAGE 2 — Build the network

```
keerthi@KeerthiVarma:~$ sudo ip netns add host1
sudo ip netns add router
sudo ip netns add host2
keerthi@KeerthiVarma:~$ sudo ip netns add host1
sudo ip netns add router
sudo ip netns add host2
Cannot create namespace file "/run/netns/host1": File exists
Cannot create namespace file "/run/netns/router": File exists
Cannot create namespace file "/run/netns/host2": File exists
keerthi@KeerthiVarma:~$ sudo ip link add veth-h1 type veth peer name veth-r1
sudo ip link add veth-r2 type veth peer name veth-h2
keerthi@KeerthiVarma:~$ sudo ip link set veth-h1 netns host1
sudo ip link set veth-r1 netns router
sudo ip link set veth-r2 netns router
sudo ip link set veth-h2 netns host2
```

STAGE 3 — Configure IP addresses (Note: In terminal 1 Only)

```
keerthi@KeerthiVarma:~$ sudo ip netns exec host1 ip addr add 10.0.1.1/30 dev veth-h1
sudo ip netns exec host1 ip link set veth-h1 up
sudo ip netns exec host1 ip link set lo up
keerthi@KeerthiVarma:~$ sudo ip netns exec router ip addr add 10.0.1.2/30 dev veth-r1
sudo ip netns exec router ip addr add 10.0.2.1/30 dev veth-r2
sudo ip netns exec router ip link set veth-r1 up
sudo ip netns exec router ip link set veth-r2 up
sudo ip netns exec router ip link set lo up
keerthi@KeerthiVarma:~$ sudo ip netns exec host2 ip addr add 10.0.2.2/30 dev veth-h2
sudo ip netns exec host2 ip link set veth-h2 up
sudo ip netns exec host2 ip link set lo up
```

STAGE 4 — Enable routing

```
keerthi@KeerthiVarma:~$ sudo ip netns exec router sysctl -w net.ipv4.ip_forward=1
sudo ip netns exec host1 ip route add 10.0.2.0/30 via 10.0.1.2
sudo ip netns exec host2 ip route add 10.0.1.0/30 via 10.0.2.1
net.ipv4.ip_forward = 1
```

STAGE 5 — Test connectivity

```
keerthi@KeerthiVarma:~$ sudo ip netns exec host1 ping -c 3 10.0.2.2
PING 10.0.2.2 (10.0.2.2) 56(84) bytes of data.
64 bytes from 10.0.2.2: icmp_seq=1 ttl=63 time=2.83 ms
64 bytes from 10.0.2.2: icmp_seq=2 ttl=63 time=0.096 ms
64 bytes from 10.0.2.2: icmp_seq=3 ttl=63 time=0.165 ms

--- 10.0.2.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2013ms
```

STAGE 6 — Add delay and loss

```
keerthi@KeerthiVarma:~$ sudo ip netns exec router tc qdisc add veth-r2 root netem delay 20ms loss 1%
Unknown qdisc "veth-r2", hence option "root" is unparsable
keerthi@KeerthiVarma:~$ sudo ip netns exec router tc qdisc show dev veth-r2
qdisc noqueue 0: root refcnt 2
```

STAGE 8 — Start three captures simultaneously

Terminal 1 — capture at host1

```
keerthi@KeerthiVarma:~$ sudo ip netns exec host1 tshark -i veth-h1 -w /tmp/cap1.pcap
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-h1'
```

Terminal 2 — capture at router

```
keerthi@KeerthiVarma:~$ sudo ip netns exec router tshark -i veth-r1 -w /tmp/cap2.pcap
[sudo: authenticate] Password:
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-r1'
```

Terminal 3 — capture at host2

```
keerthi@KeerthiVarma:~$ sudo ip netns exec host2 tshark -i veth-h2 -w /tmp/cap3.pcap
[sudo: authenticate] Password:
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-h2'
```

STAGE 9 — Send real message from host1 to host2

Terminal 4 - First start the receiver on host2

```
keerthi@KeerthiVarma:~$ sudo ip netns exec host2 nc -l 1234
[sudo: authenticate] Password:
^Z
[1]+  Stopped                 sudo ip netns exec host2 nc -l 1234
keerthi@KeerthiVarma:~$
```

Terminal 5 - Connect from host1

```
keerthi@KeerthiVarma:~$ sudo ip netns exec host1 nc 10.0.2.2 1234
[sudo: authenticate] Password:
sudo: Authentication failed, try again.
[sudo: authenticate] Password:
Hello from host1!
This is Alekhya
Computer Networks Lab
Three Point Capture Experiment
Message traveling through router
```

Terminal 4

```
keerthi@KeerthiVarma:~$ sudo ip netns exec host2 nc -l 1234
[sudo: authenticate] Password:
^Z
[1]+  Stopped                 sudo ip netns exec host2 nc -l 1234
keerthi@KeerthiVarma:~$
keerthi@KeerthiVarma:~$
```

Terminal 3

```
keerthi@KeerthiVarma:~$ sudo ip netns exec host2 tshark -i veth-h2 -w /tmp/cap3.pcap
[sudo: authenticate] Password:
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-h2'
21
```

Terminal 2

```
keerthi@KeerthiVarma:~$ sudo ip netns exec router tshark -i veth-r1 -w /tmp/cap2.pcap
[sudo: authenticate] Password:
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-r1'
23
```

Terminal 1

```
sudo ip netns exec host1 ip link set veth-h1 up
sudo ip netns exec host1 ip link set lo up
keerthi@KeerthiVarma:~$ sudo ip netns exec router ip addr add 10.0.1.2/30 dev veth-r1
sudo ip netns exec router ip addr add 10.0.2.1/30 dev veth-r2
sudo ip netns exec router ip link set veth-r1 up
sudo ip netns exec router ip link set veth-r2 up
sudo ip netns exec router ip link set lo up
keerthi@KeerthiVarma:~$ sudo ip netns exec host2 ip addr add 10.0.2.2/30 dev veth-h2
sudo ip netns exec host2 ip link set veth-h2 up
sudo ip netns exec host2 ip link set lo up
keerthi@KeerthiVarma:~$ sudo ip netns exec router sysctl -w net.ipv4.ip_forward=1
sudo ip netns exec host1 ip route add 10.0.2.0/30 via 10.0.1.2
sudo ip netns exec host2 ip route add 10.0.1.0/30 via 10.0.2.1
net.ipv4.ip_forward = 1
keerthi@KeerthiVarma:~$ sudo ip netns exec host1 ping -c 3 10.0.2.2
PING 10.0.2.2 (10.0.2.2) 56(84) bytes of data.
64 bytes from 10.0.2.2: icmp_seq=1 ttl=63 time=2.83 ms
64 bytes from 10.0.2.2: icmp_seq=2 ttl=63 time=0.096 ms
64 bytes from 10.0.2.2: icmp_seq=3 ttl=63 time=0.165 ms

--- 10.0.2.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2013ms
keerthi@KeerthiVarma:~$ sudo ip netns exec router tc qdisc add veth-r2 root netem delay 20ms loss 1%
Unknown qdisc "veth-r2", hence option "root" is unparsable
keerthi@KeerthiVarma:~$ sudo ip netns exec router tc qdisc show dev veth-r2
qdisc noqueue 0: root refcnt 2
keerthi@KeerthiVarma:~$ Terminal 1 — capture at host1
Command 'Terminal' not found, but can be installed with:
sudo apt install terminal.app
keerthi@KeerthiVarma:~$ sudo ip netns exec host1 tshark -i veth-h1 -w /tmp/cap1.pcap
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-h1'
23
```

STAGE 12 — Fix permissions

```
keerthi@KeerthiVarma:~$ sudo ip netns exec host2 nc -l 1234
[sudo: authenticate] Password:
^Z
[1]+  Stopped                 sudo ip netns exec host2 nc -l 1234
keerthi@KeerthiVarma:~$
keerthi@KeerthiVarma:~$ sudo chmod 644 /tmp/cap1.pcap /tmp/cap2.pcap /tmp/cap3.pcap
keerthi@KeerthiVarma:~$
```

STAGE 13 — Check TTL

```
keerthi@KeerthiVarma:~$ sudo chmod 644 /tmp/cap1.pcap /tmp/cap2.pcap /tmp/cap3.pcap
keerthi@KeerthiVarma:~$ tshark -r /tmp/cap1.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
tshark -r /tmp/cap2.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
tshark -r /tmp/cap3.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
      5 63
      6 64
      5 63
      6 64
      6 63
      5 64
keerthi@KeerthiVarma:~$
```

STAGE 14 — Read your actual message from the capture

```
keerthi@KeerthiVarma:~$ sudo apt install -y tshark iproute2 netcat-openbsd wireshark
[sudo: authenticate] Password:
tshark is already the newest version (4.6.4-1).
iproute2 is already the newest version (6.19.0-1ubuntu1.1).
iproute2 set to manually installed.
netcat-openbsd is already the newest version (1.234-1).
wireshark is already the newest version (4.6.4-1).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 74
```

Create 3 nodes:

```
keerthi@KeerthiVarma:~$ sudo apt install -y tshark iproute2 netcat-openbsd wireshark
[sudo: authenticate] Password:
tshark is already the newest version (4.6.4-1).
iproute2 is already the newest version (6.19.0-1ubuntu1.1).
iproute2 set to manually installed.
netcat-openbsd is already the newest version (1.234-1).
wireshark is already the newest version (4.6.4-1).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 74
keerthi@KeerthiVarma:~$ sudo ip netns add host1
sudo ip netns add router
sudo ip netns add host2
Cannot create namespace file "/run/netns/host1": File exists
Cannot create namespace file "/run/netns/router": File exists
Cannot create namespace file "/run/netns/host2": File exists
keerthi@KeerthiVarma:~$
```

Create cables:

```
keerthi@KeerthiVarma:~$ sudo ip link add veth-h1 type veth peer name veth-r1
sudo ip link add veth-r2 type veth peer name veth-h2
keerthi@KeerthiVarma:~$ sudo ip link set veth-h1 netns host1
sudo ip link set veth-r1 netns router
sudo ip link set veth-r2 netns router
sudo ip link set veth-h2 netns host2
Error: An interface with the same name exists in the target netns.
Error: An interface with the same name exists in the target netns.
Error: An interface with the same name exists in the target netns.
Error: An interface with the same name exists in the target netns.
keerthi@KeerthiVarma:~$ sudo ip netns exec host1 ip addr add 10.0.1.1/30 dev veth-h1
sudo ip netns exec host1 ip link set veth-h1 up
sudo ip netns exec host1 ip link set lo up
Error: ipv4: Address already assigned.
keerthi@KeerthiVarma:~$ sudo ip netns exec router ip addr add 10.0.1.2/30 dev veth-r1
sudo ip netns exec router ip addr add 10.0.2.1/30 dev veth-r2
sudo ip netns exec router ip link set veth-r1 up
sudo ip netns exec router ip link set veth-r2 up
sudo ip netns exec router ip link set lo up
Error: ipv4: Address already assigned.
Error: ipv4: Address already assigned.
keerthi@KeerthiVarma:~$ sudo ip netns exec host2 ip addr add 10.0.2.2/30 dev veth-h2
sudo ip netns exec host2 ip link set veth-h2 up
sudo ip netns exec host2 ip link set lo up
Error: ipv4: Address already assigned.
keerthi@KeerthiVarma:~$ sudo ip netns exec router sysctl -w net.ipv4.ip_forward=1
sudo ip netns exec host1 ip route add 10.0.2.0/30 via 10.0.1.2
sudo ip netns exec host2 ip route add 10.0.1.0/30 via 10.0.2.1
net.ipv4.ip_forward = 1
RTNETLINK answers: File exists
RTNETLINK answers: File exists
keerthi@KeerthiVarma:~$ sudo ip netns exec host1 ping -c 3 10.0.2.2
PING 10.0.2.2 (10.0.2.2) 56(84) bytes of data.
```

```
keerthi@KeerthiVarma:~$ sudo ip netns exec host1 ping -c 3 10.0.2.2
PING 10.0.2.2 (10.0.2.2) 56(84) bytes of data.
64 bytes from 10.0.2.2: icmp_seq=1 ttl=63 time=1.64 ms
64 bytes from 10.0.2.2: icmp_seq=2 ttl=63 time=0.301 ms
64 bytes from 10.0.2.2: icmp_seq=3 ttl=63 time=0.325 ms

--- 10.0.2.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2032ms
rtt min/avg/max/mdev = 0.301/0.755/1.640/0.625 ms
keerthi@KeerthiVarma:~$
```

Add delay and loss:

```
keerthi@KeerthiVarma:~$ sudo ip netns exec router tc qdisc add dev veth-r2 root netem delay 20ms loss
1%
Command line is not complete. Try option "help"
1%: command not found
keerthi@KeerthiVarma:~$ sudo ip netns exec router tc qdisc show dev veth-r2
qdisc noqueue 0: root refcnt 2
```

TERMINAL 1 — Capture at host1

```
keerthi@KeerthiVarma:~$ sudo ip netns exec router tc qdisc show dev veth-r2
qdisc noqueue 0: root refcnt 2
keerthi@KeerthiVarma:~$ sudo ip netns exec host1 tshark -i veth-h1 -w /tmp/cap1.pcap
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-h1'
```

TERMINAL 2 — Capture at router

```
keerthi@KeerthiVarma:~$ sudo ip netns exec router tshark -i veth-r1 -w /tmp/cap2.pcap
[sudo: authenticate] Password:
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-r1'
```

TERMINAL 3 — Capture at host2

```
keerthi@KeerthiVarma:~$ sudo ip netns exec host2 tshark -i veth-h2 -w /tmp/cap3.pcap
[sudo: authenticate] Password:
Running as user "root" and group "root". This could be dangerous.
Capturing on 'veth-h2'
```

```
keerthi@KeerthiVarma:~$ sudo ip netns exec host1 nc 10.0.2.2 1234
[sudo: authenticate] Password:
Hello from host1!
This is Keerthi
Three Point Capture Lab
Message going through router
^C
keerthi@KeerthiVarma:~$ sudo ip netns exec host1 nc 10.0.2.2 1234
Hello from host1!
This is Keerthi
Three Point Capture Lab
Message going through router
```

Step 6 — Press Ctrl+C in all terminals

TERMINAL 4 — Analyse results

Fix permissions:

```
keerthi@KeerthiVarma:~$ sudo chmod 644 /tmp/cap1.pcap /tmp/cap2.pcap /tmp/cap3.pcap
keerthi@KeerthiVarma:~$
```

Check files are not empty:

```
keerthi@KeerthiVarma:~$ sudo chmod 644 /tmp/cap1.pcap /tmp/cap2.pcap /tmp/cap3.pcap
keerthi@KeerthiVarma:~$ ls -lh /tmp/cap1.pcap /tmp/cap2.pcap /tmp/cap3.pcap
-rw-r--r-- 1 root root 5.1K Nov  9 16:52 /tmp/cap1.pcap
-rw-r--r-- 1 root root 5.1K Nov  9 16:51 /tmp/cap2.pcap
-rw-r--r-- 1 root root 4.7K Nov  9 16:51 /tmp/cap3.pcap
keerthi@KeerthiVarma:~$
```

Check TTL at all 3 points:

```
keerthi@KeerthiVarma:~$ sudo chmod 644 /tmp/cap1.pcap /tmp/cap2.pcap /tmp/cap3.pcap
keerthi@KeerthiVarma:~$ ls -lh /tmp/cap1.pcap /tmp/cap2.pcap /tmp/cap3.pcap
-rw-r--r-- 1 root root 5.1K Nov  9 16:52 /tmp/cap1.pcap
-rw-r--r-- 1 root root 5.1K Nov  9 16:51 /tmp/cap2.pcap
-rw-r--r-- 1 root root 4.7K Nov  9 16:51 /tmp/cap3.pcap
keerthi@KeerthiVarma:~$ tshark -r /tmp/cap1.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
tshark -r /tmp/cap2.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
tshark -r /tmp/cap3.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
     13 63
     15 64
     13 63
     15 64
     15 63
     13 64
keerthi@KeerthiVarma:~$
```

Read your actual message from router capture:

```
keerthi@KeerthiVarma:~$ tshark -r /tmp/cap2.pcap -Y "tcp.port == 1234" -T fields -e tcp.payload
2>/dev/null | xxd | head -20




48656c6c6f2066726f6d20686f737431210a

546869732069732046565727468690a

5468726565520506f696e742043617074757265204c61620a

4d65737361676520676f696e67207468726f75676820726f757465720a




48656c6c6f2066726f6d20686f737431210a

546869732069732046565727468690a

5468726565520506f696e742043617074757265204c61620a

4d65737361676520676f696e67207468726f75676820726f757465720a
```

Check delay:

```
keerthi@KeerthiVarma:~$ tshark -r /tmp/cap1.pcap -Y "tcp.flags.syn==1 && tcp.flags.ack==0" -T
fields -e frame.time_relative 2>/dev/null
tshark -r /tmp/cap2.pcap -Y "tcp.flags.syn==1 && tcp.flags.ack==0" -T
fields -e frame.time_relative 2>/dev/null
tshark -r /tmp/cap3.pcap -Y "tcp.flags.syn==1 && tcp.flags.ack==0" -T
fields -e frame.time_relative 2>/dev/null
tshark: option requires an argument: T

Usage: tshark [options] ...

Capture interface:
  -i <interface>, --interface <interface>
                           name or idx of interface (def: first non-loopback)
  -f <capture filter>      packet filter in libpcap filter syntax
  -s <snaplen>, --snapshot-length <snaplen>
                           packet snapshot length (def: appropriate maximum)
  -p, --no-promiscuous-mode
                           don't capture in promiscuous mode
  -I, --monitor-mode       capture in monitor mode, if available
  -B <buffer size>, --buffer-size <buffer size>
                           size of kernel buffer in MiB (def: 2MiB)
  -y <link type>, --linktype <link type>
                           link layer type (def: first appropriate)
  --time-stamp-type <type> timestamp method for interface
  -D, --list-interfaces    print list of interfaces and exit
  -L, --list-data-link-types
                           print list of link-layer types of iface and exit
  --list-time-stamp-types  print list of timestamp types for iface and exit
```

Check retransmissions:

```
keerthi@KeerthiVarma:~$ tshark -r /tmp/cap1.pcap -Y "tcp.analysis.retransmission" 2>/dev/null | wc -l
tshark -r /tmp/cap2.pcap -Y "tcp.analysis.retransmission" 2>/dev/null | wc -l
tshark -r /tmp/cap3.pcap -Y "tcp.analysis.retransmission" 2>/dev/null | wc -l
0
0
0
```

Check RTT:

```
keerthi@KeerthiVarma:~$ sudo ip netns exec host1 mtr --report 10.0.2.2
Start: 2026-11-09T16:57:49+0000
HOST: KeerthiVarma                Loss%   Snt   Last   Avg  Best  Wrst StDev
  1.|-- 10.0.1.2                   0.0%    10    0.1   0.7   0.1   5.3   1.6
  2.|-- 10.0.2.2                   0.0%    10    0.2   0.2   0.1   1.1   0.3
keerthi@KeerthiVarma:~$
```

TERMINAL 4 — Open Wireshark

```
  --log-fatal-domains <list>
                           list of domains that cause the program to abort
  --log-debug <[!]list>    list of domains with "debug" level
  --log-noisy <[!]list>    list of domains with "noisy" level
  --log-file <path>        file to output messages to (in addition to stderr)

Miscellaneous:
  -h, --help               display this help and exit
  -v, --version            display version info and exit
  -o <name>:<value> ...    override preference setting
  -K <keytab>              keytab file to use for kerberos decryp
  -G [report]              dump one of several available reports
                           default report="fields"
                           use "-G help" for more help

Dumpcap can benefit from an enabled BPF JIT compiler if available
You might want to enable it by executing:
 "echo 1 > /proc/sys/net/core/bpf_jit_enable"
Note that this can make your system less secure!
keerthi@KeerthiVarma:~$ tshark -r /tmp/cap1.pcap -Y "tcp.analysis
tshark -r /tmp/cap2.pcap -Y "tcp.analysis.retransmission" 2>/dev/
tshark -r /tmp/cap3.pcap -Y "tcp.analysis.retransmission" 2>/dev/
0
0
0
keerthi@KeerthiVarma:~$ sudo ip netns exec host1 mtr --report 10.
Start: 2026-11-09T16:57:49+0000
HOST: KeerthiVarma                Loss%   Snt   Last   Avg  Best  Wrst StDev
  1.|-- 10.0.1.2                   0.0%    10    0.1   0.7   0.1   5.3   1.6
  2.|-- 10.0.2.2                   0.0%    10    0.2   0.2   0.1   1.1   0.3
keerthi@KeerthiVarma:~$ wireshark /tmp/cap2.pcap &
[3] 4878
keerthi@KeerthiVarma:~$
```

